

DHANANJAI PRATAP SINGH

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Education

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology in Computer Science Engineering- Specialization in Internet Of Things

Sep 2022 – Sep 2026

CGPA : 8.5

Kendriya Vidyalaya No-2, KPA ,West Bengal

Intermediate (12th PCM)

July 2020 – July 2021

93%

Experience

Explorin

Full Stack Intern

July 2025 – August 2025

Moradabad, Uttar Pradesh

- Developed modular SEO-optimized Next.js + TypeScript interfaces and optimized GraphQL/Prisma APIs, designed DB for high-performance, scalability and to reduce latency. Took end-to-end ownership of development to deploy, managed CI/CD, coordinated with backend teams for integration testing, and delivered reliable, production-ready features.

Projects

Smart Dukan | *Next.js, TypeScript, IoT Hardware, Machine Learning, IoT Cloud*

[Live Link](#)

- Developed an end-to-end **Smart Retail Ecosystem** integrating **IoT sensors** (RFID Readers / Tags, Load Cells, DHT11, ESP8266 / ESP32) and **ESP32-CAM** to eliminate slow checkout, product misplacement, and long queues .
- Engineered a lightweight computer-vision inference pipeline using **MobileNet SSD** (achieving **81.4% accuracy**) for real-time product detection, optimized for **low-latency inference** on edge devices.
- Architected a **real-time synchronization layer** using **WebSockets** to connect the IoT hardware with a Next.js dashboard for **instant cart updates**, automated billing workflows with invoice generation and AI recommendations.
- Awarded **2nd Prize** at TechXthone **Hackathon** (MIT Moradabad) and authored a **Best Paper** at the **International IIR 5.0 Conference**, for innovative contributions to AI-driven retail automation and IoT-edge synchronization.

Rail Booking System | *Next.js, PostgreSQL, Prisma, Redis, Stripe, Tailwind CSS, Twilio*

[Live Link](#)

- Developed a scalable IRCTC-inspired ticketing Booking platform enabling users to search trains, check seat availability, book tickets, track trains and manage bookings using **Next.js 14** and **PostgreSQL**, with optimized state management to support real-time seat availability and concurrent booking operations across multiple trains with **Stripe** for secure asynchronous payment.
- Engineered a **distributed locking and cleanup system** using **Redis (QStash)** to auto-rollback pending bookings after 10 mins, effectively preventing seat inventory leakage and ensuring database consistency.
- Implemented **Prisma Transactions** for seat allocation ensures **ACID** compliance, preventing double-booking and race conditions.
- Implemented additional features including multi-language translation support, pet travel booking, admin complaint management, , **Twilio** for automated PNR status alerts and PDF ticket generation.

Job Dekho | *NextJS, TypeScript, Prisma, Radix UI, Tailwind, MongoDB*

[Live Link](#)

- Developed a job portal application with **modular** architecture, **reusable** components, and efficient state management using **Context API** and integrated secure authentication using **JWT** for encrypted user identity and role-based authorization,
- Implemented **job** and **company** management with features like applications, reviews, search, filtering, apply and **CRUD operations** via API endpoints. Integrated company hiring and job-seeker workflows in a unified platform.

Mini - Amazon | *ReactJS, Tailwind CSS, Dummy JSON API, Redux*

[Live Link](#)

- Built a responsive pixel perfect Amazon with **200+** products using React JS, Tailwind CSS, and React Router, featuring real-time search, filtering, secure authentication, and dynamic order management.
- Implemented product detail navigation, real-time search, and a fully functional 'Add to Cart' feature. Integrated APIs and used **Redux** for state management, enhancing performance and UX.

Technical Skills

Languages: C++, C, Python, Java, TypeScript

Web Technologies: HTML, CSS, JavaScript , GraphQL, REST APIs, Responsive Design, SEO Optimization.

Frameworks & Libraries: React, Next.js ,Vue.js , FastAPI, Prisma, Radix UI, Tailwind CSS.

Hardware / IoT: IoT Sensors, ESP32, ESP8266, Blynk, ESP32-CAM, Arduino UNO, Arduino Nano, Arduino IDE.

Databases: MySQL, PostgreSQL, MongoDB, SQL.

Developer Tools: Git, GitHub, VS Code, Postman.

Others: SDLC, Generative AI, Operating Systems, Database , Microservices, Solid Principles, DB Design.

Awards and Achievements

- Qualified **GATE DA 2026 (AIR 6,358)**, **GATE DA 2025 (AIR 7,747)**, and **GATE CS 2026 (AIR 20,661)**.
- Secured **2nd place** out of **300+** participants at Hi-Tech **TechXthone, Hackathon** held at MIT Moradabad.
- Best Research Paper Award** on "Connected Retail With Smart Shelves and Inventory Management" at International IoT, Robotics and Automation 5.0 Conference (2025), among **290+** Research papers.
- Secured a position in the **Top 10** in a **state-level hackathon** at SRMSCET, Bareilly
- Awarded **Academic Achiever Scholarship** of 10,000 INR by the college.
- Solved **800+** Data Structures & Algorithms problems across platforms including LeetCode, GeeksforGeeks, HackerRank.

Extracurriculars

- Led my college football team in the **University Games**, and together we secured the **1st** position.